Distinct Values Subsequences

Time limit: 1.00 s
Memory limit: 512 MB

CSES 3421 | Sorting and Searching | https://cses.fi/problemset/task/3421

Given an array of n integers, count the number of subsequences where each element is distinct.

A subsequence is a sequence of array elements from left to right that may have gaps.

Input

The first line has an integer n: the array size.

The second line has n integers $x_1,x_2,\ldots,x_n$: the array contents.

Output

Print the number of subsequences with distinct elements. The answer can be large, so print it modulo 10^9+7.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le x_i \le 10^9$

Example

Input
4 1 2 1 3

Output
11

Explanation: The subsequences are [1] (two times), [2], [3], [1,2], [1,3] (two times), [2,1], [2,3], [1,2,3] and [2,1,3].